Recall...

How to find the general solution of a linear system...

Example. Solve $\begin{cases} -x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - 4x_2 = -4 \\ -x_1 + 2x_2 + 3x_3 = -1 \end{cases}$

1. Given a linear system, we first write its augmented matrix.

$$\begin{cases} -x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - 4x_2 = -4 \\ -x_1 + 2x_2 + 3x_3 = -1 \end{cases} \rightsquigarrow \left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{array}\right]$$

2. Using elementary row operations we find the R.E.F. of the augmented matrix.

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{ccc|c} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$ R.E.F. (done last time)

Recall that for any augmented matrix ...

- the location of a leading 1 in its R.E.F. is called a *pivot position*.
- a column containing a pivot position is called a *pivot column*.
- a variable corresponding to a pivot column is called a *basic* variable.
- a variable corresponding to a non-pivot column is called a *free* variable.

Recall (cont.)

How to find the general solution of a linear system...

Example. Solve $\begin{cases} -x_1 & +2x_2 & +x_3 & = 1 \\ 2x_1 & -4x_2 & & = -4 \\ -x_1 & +2x_2 & +3x_3 & = -1 \end{cases}$

3. We write the linear system whose augmented matrix is the R.E.F. above.

$$\left[\begin{array}{ccc|c} \mathbf{1} & -2 & 0 & -2 \\ 0 & 0 & \mathbf{1} & -1 \\ 0 & 0 & 0 & 0 \end{array}\right] \text{ R.E.F. } \rightsquigarrow \begin{cases} x_1 & -2x_2 & & = -2 \\ & & x_3 & = -1 \\ & & 0 & = 0 \end{cases}$$

x_1 x_2 x_3

- x_1 and x_3 are **basic** variables.
- x_2 is a **free** variable.

The general solution is : $\begin{cases} x_1 = 2x_2 - 2 \\ x_2 \text{ is free} \\ x_3 = -1 \end{cases}$

To check the answer:

$$\begin{cases} -(2x_2 - 2) & +2x_2 & -1 & = 1 \quad \checkmark \\ 2(2x_2 - 2) & -4x_2 & & = -4 \quad \checkmark \\ -(2x_2 - 2) & +2x_2 & +3(-1) & = -1 \checkmark \end{cases}$$

Example

Solve the linear system $\begin{cases} x_1 + x_2 = 1 \\ x_2 + x_3 = 2 \\ x_1 + 2x_2 + x_3 = 0 \end{cases}$

Solution. Write the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 0 \end{array}\right]$$

Now we find the R.E.F.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & -1 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & -3 \end{array}\right] \text{ E.F. } \xrightarrow{R_3 \to -\frac{1}{3}R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_1 \to R_1 - R_3]{R_2 \to R_2 - 2R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \text{ R.E.F.}$$

The system is **inconsistent**.

Example

Solve the linear system $\begin{cases} 2x_1 + 2x_2 - x_4 - 1 = x_1 \\ 2x_2 + 2 = x_3 - x_1 - x_4 \\ 2x_3 = x_1 + 2x_2 + 3x_4 + 5 \end{cases}$

Solution.

First we write the system in the standard form:

$$\begin{cases} x_1 + 2x_2 - x_4 = 1 \\ x_1 + 2x_2 - x_3 + x_4 = -2 \\ -x_1 - 2x_2 + 2x_3 - 3x_4 = 5 \end{cases}$$

The augmented matrix is: $\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 1 & 2 & -1 & 1 & -2 \\ -1 & -2 & 2 & -3 & 5 \end{array}\right]$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 1 & 2 & -1 & 1 & -2 \\ -1 & -2 & 2 & -3 & 5 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & -1 & 2 & -3 \\ 0 & 0 & 2 & -4 & 6 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_2}$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & -1 & 2 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \text{ E.F. } \xrightarrow{R_2 \to -R_2} \left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \text{ R.E.F.}$$

Example

Solve the linear system $\begin{cases} 2x_1 + 2x_2 - x_4 - 1 = x_1 \\ 2x_2 + 2 = x_3 - x_1 - x_4 \\ 2x_3 = x_1 + 2x_2 + 3x_4 + 5 \end{cases}$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \rightsquigarrow \begin{cases} x_1 + 2x_2 - x_4 = 1 \\ x_3 - 2x_4 = 3 \\ 0 = 0 \end{cases}$$

x_1 x_2 x_3 x_4

x_1, x_3 are basic. x_2, x_4 are free. The general solution is :

$$\begin{cases} x_1 = -2x_2 + x_4 + 1 \\ x_2 : \text{ free} \\ x_3 = 2x_4 + 3 \\ x_4 : \text{ free} \end{cases}$$

Exercise. Check this solution.

Example

Find all values of the parameter h such that the matrix

$$\left[\begin{array}{cc|c} 1 & 3 & h \\ 2 & 6 & 1 \end{array}\right]$$

is the augmented matrix of a consistent linear system.

Solution. We try to calculate the E.F.

$$\left[\begin{array}{cc|c} 1 & 3 & h \\ 2 & 6 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 3 & h \\ 0 & 0 & 1-2h \end{array}\right]$$

- The system is inconsistent if $1 - 2h \neq 0$, i.e., $h \neq \frac{1}{2}$.

- The system is consistent if $h = \frac{1}{2}$. In this case the system has infinitely many solutions (because the second column is not a pivot column).

Example

Find all values of the parameter h such that the matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & -1 \\ 2 & h & -4 & 0 \\ -1 & -1 & 0 & 1 \end{array}\right]$$

is the augmented matrix of a consistent linear system.

Solution. We try to calculate the E.F.

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & -1 \\ 2 & h & -4 & 0 \\ -1 & -1 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & -2 & -1 \\ 0 & h-2 & 0 & 2 \\ 0 & 0 & -2 & 0 \end{array}\right]$$

- If $h - 2 = 0$ then the system is <u>inconsistent</u> because the rightmost column is a pivot column. (In this case to obtain an E.F., we need to interchange R_2 and R_3:

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & -1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & -2 & 0 \end{array}\right] \xrightarrow{R_2 \leftrightsquigarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & -2 & -1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 2 \end{array}\right]$$

- If $h - 2 \neq 0$ then the matrix is in E.F. and the system is <u>consistent</u> because the rightmost column is not a pivot column.

- Therefore the system is consistent if and only if $h \neq 2$.

Example

For which values of a and b does the linear system

$$\begin{cases} x_1 + 2x_2 &= 2 \\ 3x_1 + ax_2 &= b \end{cases}$$

have:
- a unique solution?
- infinitely many solutions?
- no solutions?

Solution. The augmented matrix is $\left[\begin{array}{cc|c} 1 & 2 & 2 \\ 3 & a & b \end{array}\right]$.

We try to calculate the E.F.

$$\left[\begin{array}{cc|c} 1 & 2 & 2 \\ 3 & a & b \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & a-6 & b-6 \end{array}\right]$$

- $\underline{a = 6}$ and $\underline{b \neq 6}$ $\Rightarrow$ the system is inconsistent (because the rightmost column is a pivot column).

- $\underline{a = 6}$ and $\underline{b = 6}$ $\Rightarrow$ the system has infinitely many solutions (because the second column is not a pivot column).

- $\underline{a \neq 6}$ $\Rightarrow$ the system has a unique solution (because the first and the second column are both pivot columns).

Recall the notions of augmented and coefficient matrix.

$$\begin{cases} -x_1 & +2x_2 & +x_3 & = 1 \\ 2x_1 & -4x_2 & & = -4 \\ -x_1 & +2x_2 & +3x_3 & = -1 \end{cases}$$

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{array}\right] \qquad \left[\begin{array}{ccc} -1 & 2 & 1 \\ 2 & -4 & 0 \\ -1 & 2 & 3 \end{array}\right]$$

augmented matrix coefficient matrix

Example

Suppose a linear system has 4 equations and 5 variables. If the R.E.F. of the coefficient matrix has 4 pivot positions the system has . . .

(A) no solutions.

(B) infinitely many solutions.

(C) a unique solution.

Solution.

• The coefficient matrix has 4 rows and 5 columns.

• 4 rows and 4 pivots ⇒ Every row of the R.E.F. of the coefficient matrix must contain a pivot position.

• Therefore the system is consistent.

• 4 pivots and 5 variables ⇒ one variable must be a free variable.

• Therefore the system has infinitely many solutions.

The correct answer is : **B**.

$$\left[\begin{array}{ccccc|c} 1 & 0 & * & 0 & 0 & * \\ 0 & 1 & * & 0 & 0 & * \\ 0 & 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 0 & 1 & * \end{array}\right]$$

An example of a R.E.F. satisfying the assumptions of the above problem

Example

Suppose we have a system of 18 equations and 27 variables. Suppose that the R.E.F. of the augmented matrix has 13 pivot columns.
Which of the following is definitely wrong?

(A) The system may have no solutions.

(B) The system may have infinitely many solutions.

(C) The system may have a unique solution.

(A) can be right : if the rightmost column is a pivot column.

(B) can be right : if the rightmost column is not a pivot column, then the system is consistent. We have 27 variables and 13 pivot columns, and $27 > 13$, therefore some variables must be free.

The correct answer is **(C)**.

for an $m \times n$ matrix (i.e., a matrix with m rows and n columns) :

- There are at most $\min\{m, n\}$ pivot positions (and pivot columns).
- If there are less than m pivot positions, then there is a row without a pivot position.
- If there are less than n positions, then there is a column which is not a pivot column.

For a system of m equations in n variables, the coefficient matrix is $m \times n$ and the augmented matrix is $m \times (n + 1)$.

Given a system of m equations in n variables...

- If the rightmost column of the augmented matrix is a pivot column, then the system is inconsistent.
- If the coefficient matrix has less than n pivots, then the system has either no solutions or infinitely many solutions.
- If a linear system is consistent and has more variables than equations (i.e., $n > m$) then it has infinitely many solutions.

Vectors

vectors

A matrix with only one column is called a vector or a column vector.

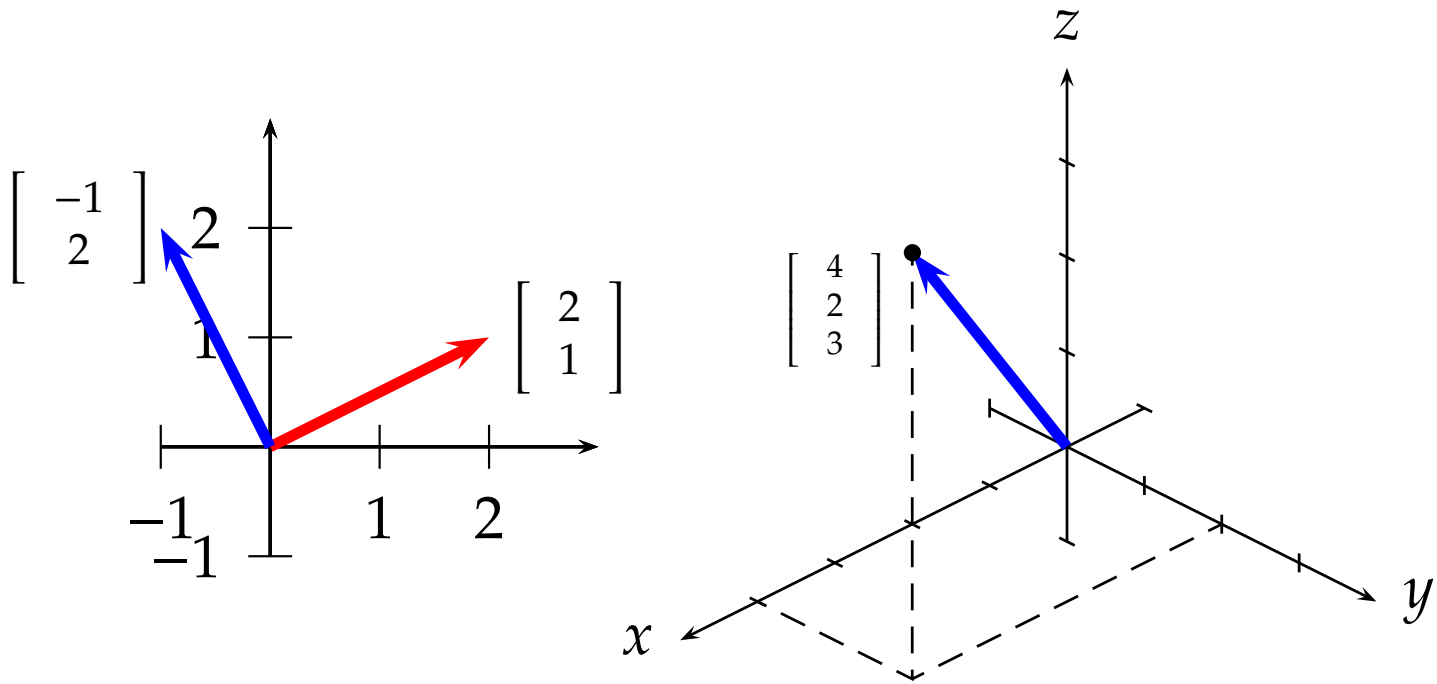
Example

$\begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \\ 2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 3 \end{bmatrix}$ are some examples of column vectors.

- We usually represent vectors by boldface letters (e.g., $\mathbf{a}, \mathbf{b}, \mathbf{u}, \mathbf{v}, \ldots$).

Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$

Vectors can be realized geometrically by arrows :



In general, points of $\mathbb{R}^n$ are represented by $n \times 1$ vectors :

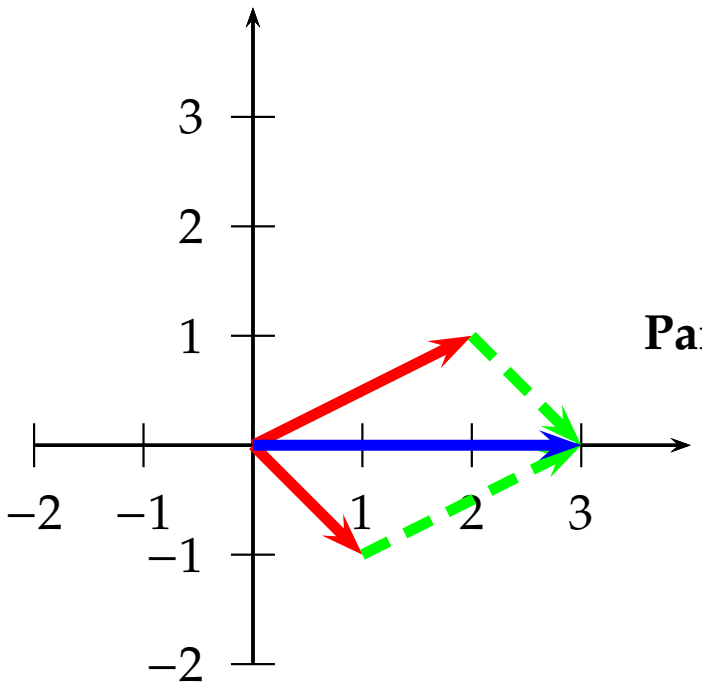
$$\begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \\ -2 \end{bmatrix} \text{ is a vector in } \mathbb{R}^5.$$

Vector addition

Given two vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ (or $\mathbb{R}^n$), we can take their sum:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \implies \mathbf{u} + \mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \implies \mathbf{u} + \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

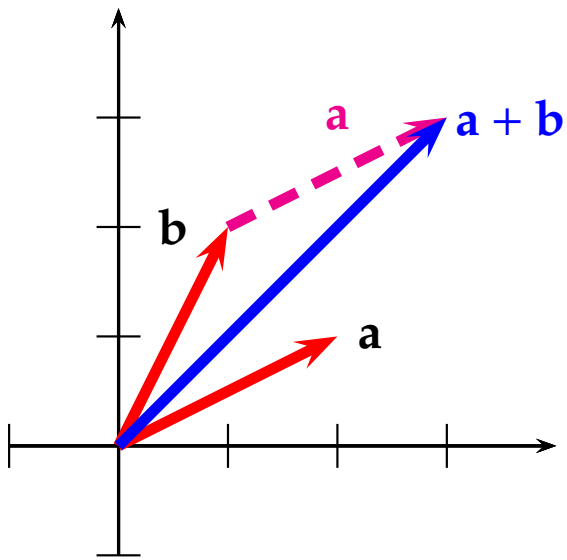


Parallelogram Law

Parallelogram Law

To find the sum of two vectors **a** and **b** geometrically, you can

1. form a parallelogram whose two sides are **a** and **b**. The sum of the vectors is determined by the corner opposite to the origin.
2. alternatively, you can draw a vector identical to **a** which originates from the tip of **b**.

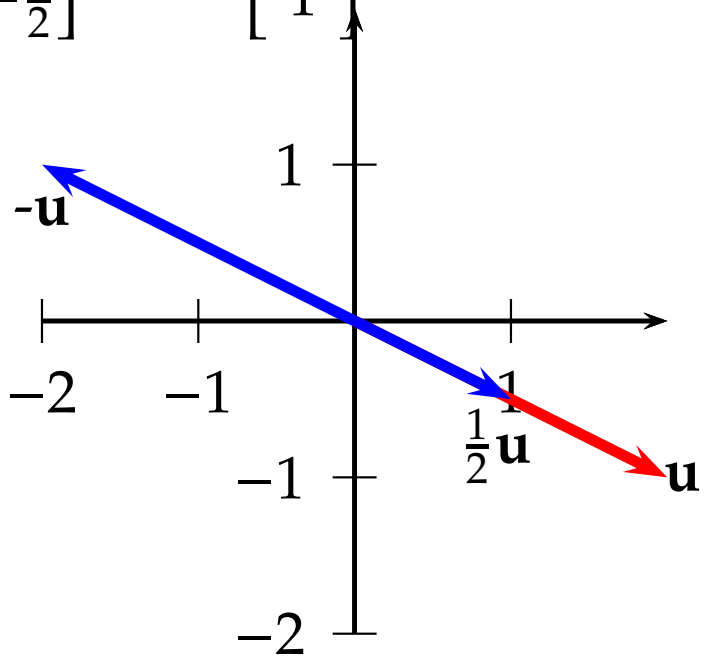


Scalar multiplication

We can also rescale vectors by any scalar.

$$2\begin{bmatrix} 1 \\ 3 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 0 \\ -2 \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \tfrac{1}{2}\mathbf{u} = \begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix} \quad -\mathbf{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$



$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$2\mathbf{u} + \mathbf{v} = 2\begin{bmatrix} 2 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

